PRACTICE PROGRAMS

Write a Java program to declare an array of integers and initialize it with the numbers 1 to 5. Then, print out the first and last element of the array.

```java
public class Main {
    public static void main(String[] args) {
        // Declare and initialize an array of integers
        int[] numbers = {1, 2, 3, 4, 5};

        // Print the first element of the array
        System.out.println("First element: " + numbers[0]);

        // Print the last element of the array
        // Since array indices start at 0, the last element is at length-1
        System.out.println("Last element: " + numbers[numbers.length - 1]);
    }
}
```

Explain with a Java program why arrays in Java have a fixed size. Try to add a new element to an already full array and catch the ArrayIndexOutOfBoundsException that is thrown.

```
public class Main {
    public static void main(String[] args) {
        // Declare and initialize an array of integers with 5 elements
        int[] myArray = {1, 2, 3, 4, 5};

        // Attempt to add a new element at index 5
        // This will cause an ArrayIndexOutOfBoundsException,
terminating the program
        myArray[5] = 6;

        // This line will not be executed
        System.out.println("This line will not be printed.");
    }
}
```

Write a Java program that takes two numbers as input and calculates their sum, difference, product, quotient, and remainder, then prints the results.

```java
public class ArithmeticOperations {
    public static void main(String[] args) {
        // Directly provide the two numbers
        int number1 = 15;
        int number2 = 4;

        // Calculate sum, difference, product, quotient, and remainder
        int sum = number1 + number2;
        int difference = number1 - number2;
        int product = number1 * number2;
        double quotient = number1 / (double) number2; // Cast to double for floating-point
division
        int remainder = number1 % number2;
        // Print the results
        System.out.println("Given numbers: " + number1 + " and " + number2);
        System.out.println("Sum: " + sum);
        System.out.println("Difference: " + difference);
        System.out.println("Product: " + product);
        System.out.println("Quotient: " + quotient);
        System.out.println("Remainder: " + remainder);
    }
}
```

Write a Java program to create a 2x2 string array, initialize it with values, and then print all the values in a matrix format.

```
public class Main {
    public static void main(String[] args) {
        // Create and initialize a 2x2 string array
        String[][] matrix = {
            {"A1", "A2"},
            {"B1", "B2"}
        };

        // Print the values in a matrix format
        for (int i = 0; i < matrix.length; i++) { // Loop through rows
            for (int j = 0; j < matrix[i].length; j++) { // Loop through columns
                System.out.print(matrix[i][j] + " "); // Print each element followed
by a space
            }
            System.out.println(); // Move to the next line after printing all
columns in a row
        }
    } }
```